Dylan Mills

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EDUCATION

University of Colorado, Boulder **Boulder, CO**
B.S. Mechanical Engineering | *Specialization in Control Systems* 2020 — 2025

PROFESSIONAL EXPERIENCE

QualCorr Engineering **Castle Rock, CO**
Electrical Integration & Test Engineer *June 2025 — July 2026*
- Led teams of 8 technicians in integrating and testing $3M+ of electrical systems over 75 energy and water pipelines
- Analyzed data with Excel to characterize corrosion system performance, presented to customers, saved 100+ assets
- Collected data with data loggers, LDROs, soil resistivity meters, multimeters, locators, PCMs, EMF meters, etc.
- Designed, calibrated and tested cathodic protection systems including ICCP, galvanic systems, anodes, and rectifiers
- Produced design calculations in Excel, drawings in AutoCAD, maps in Google Earth, and submittals in Revu
- Repaired and maintained field and lab test equipment such as air monitors, reference electrodes, and weld equipment
- Managed project resources and schedule to meet technical requirements, project deadlines, and industry standards

CU/Los Alamos National Laboratory (LANL) **Los Alamos, NM**
Mechatronics Engineer - Sponsored Capstone Project *September 2024 — May 2025*
- Developed 3-DOF robotic arm system (Link) with integrated sensors for precise X-ray imaging of explosive materials
- Designed embedded control system using Raspberry Pi, Arduino, ESP32, motor drivers, sensors, and CAN bus
- Developed KiCad electrical schematics and system architecture diagrams to define components and connections
- Implemented limit-switch, encoder, and IMU feedback for ±0.5 in positioning accuracy within a 68 x 56 in workspace
- Engineered wiring and component placement for multi-voltage power distribution, IP67 waterproofing, and access
- Refined designs through PDR, CDR, meetings with LANL management, and field testing with the FBI Bomb Squad
- Awarded “Distinction in Design” by the Department of Mechanical Engineering at University of Colorado Boulder

PROJECT EXPERIENCE

CU Control Systems Laboratory **Boulder, CO**
Control Systems Engineer *Spring 2025*
- Designed feedback controllers (PID, full-state feedback, LQR, Lead-Lag) to optimize system response characteristics
- Modeled dynamic systems (gyroscope, rotary double inverted pendulum, ball-on-beam, etc.) in MATLAB/Simulink
- Analyzed response, stability, and performance using transfer functions, root locus, Bode, and time-domain methods

CU Mechatronics Laboratory **Boulder, CO**
Mechatronics Engineer *Fall 2024*
- Developed autonomous robot for maze exploration, target detection, tracking, and engagement using computer vision
- Designed and developed a 360° PID-controlled tracking turret using Pixy2 camera, Arduino, motors, and a crossbow
- Programmed object memorization, color identification and distance estimation in C++ to prioritize between targets
- Integrated Raspberry Pi 4, a camera, IR sensors, and ultrasonic sensors with an omnidirectional wheel chassis
- Designed, 3D-printed, and refined custom chassis and mounting components using SolidWorks and LulzBot TAZ 6

CU Battery Powered Vehicle Competition **Boulder, CO**
Project Manager *Spring 2023*
- Led a five-engineer team to 2nd place out of 30 in a $200 design-build battery-powered vehicle competition
- Optimized vehicle frame and gear ratios for high torque and load capacity, reliably carrying 900 lbs up a 10% grade
- Modeled drivetrain and frame assemblies in SolidWorks, performed FEA, built timelines, and oversaw integration

CU BioFuel Cell Laboratory **Boulder, CO**
Test Engineer *Spring 2023*
- Evaluated glucose biofuel cell efficiency under varied gas and liquid inputs by monitoring long-term electrical output
- Fabricated polymer membranes via controlled casting/rolling of molarity-specific solutions to modify rates of reaction

CU Bio-Inspired Robotics Laboratory **Boulder, CO**
Robotics Engineer *Spring 2024*
- Designed 220 x 500 x 100 mm robot capable of identifying ultrasonic sensor signal jamming and adapting frequency
- Programmed Arduino-based control algorithms, inspired by the Big Brown Bat, for adaptive echolocation navigation

SKILLS

Software: Python, C++, MATLAB/Simulink, SolidWorks (CSWA), AutoCAD, KiCad, Arduino IDE, Revu
Technical: CAN bus, PID/LQR, FEA, CFD, Electrical Testing, Calibration, Fabrication, Metalworking, Welding

HOBBIES

Snowboarding, Chess, Reading, Traveling, Drawing, Surfing, Weightlifting, Hiking, Mountain Biking, Fishing, Exploring